

A prospectively frozen operator for the extent of a source-maintained field on a capacity-limited lattice

Emergent Dynamics Laboratory

Abstract

The extent of a field maintained by a localised source in a discrete, capacity-limited medium is set jointly by transport, mortality and the source’s own history, and a continuum reading of it is not available at the scale studied here: it errs by 18.74 % and 19.29 % against the discrete operator. Such extents are ordinarily reported after the fact, so it is rarely possible to say whether a model predicted them or accommodated them. We therefore constructed a one-step conditional operator directly from frozen kinetics, derived an equivalence margin of 2.9 % from named error terms, froze two absolute predictions and their ratio, hashed the analysis code, and only then ran 28 fresh arms. The observed medians met the frozen predictions to -0.14% and $+0.24\%$, and the mobility ratio to $+0.39\%$ with unity excluded; every interval lay wholly inside the margin, with no extinction and no invalid arm. We then asked why an earlier record had shown deficits of -1.833% and -5.172% . Over 116 historical arms the cumulative radial profile follows the operator at every one of 15 radii, maximum absolute z of 0.64, so the field was never in deficit; a surrogate with no lattice dynamics at all already carries -1.206% to -1.307% , and the remainder requires the source’s own wandering, which moves the construction from -0.6563% to -4.424% and then to -5.695% once the empirical birth flux is added. The operator is exact given the state but the marginal density equation does not close, and the qualification holds at one parameter point.

Scope statement. This paper concerns a transport-and-mortality field around a source in a lattice model. It makes no claim about biology. The following are reported unconditionally and without qualification:

```
H3_STATUS = NOT_TESTED    REPRODUCTION_STATUS = NOT_TESTED    HEREDITY_STATUS = NOT_TESTED
AUTONOMOUS_COHESION_STATUS = NOT_ESTABLISHED    X_LAWSPEC_BASELINE = UNCHANGED
ARCHITECTURE_CHANGE_NECESSITY = NOT_ESTABLISHED
```

1 The quantity, and why it is not trivial

A localised source injects particles into a medium; the particles move, and they die. At stationarity the field around the source has a characteristic extent. Writing down that extent looks like a textbook exercise: balance injection against decay, take the diffusion limit, read off a localisation length. The exercise fails here for three separate reasons, and the failures are the reason the quantity is worth measuring rather than deriving.

First, the medium is discrete and the reported extent is quantised. The observable used throughout this paper, written r_{80} , is the smallest *achievable* lattice distance at which the empirical cumulative field first reaches 0.8. It is therefore supported on the discrete set of achievable toroidal distances, not on the real line, and near the values of interest that set is coarse. A continuum expression with the same localisation length is not a good approximation to it: against the discrete

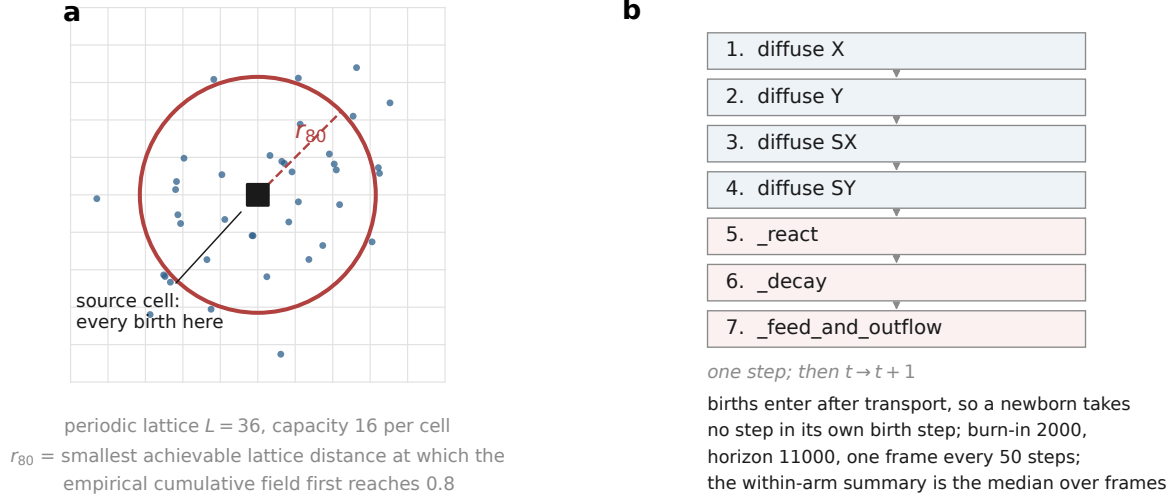


Figure 1: The model and the reported observable. (a) A point source on a periodic lattice with a per-cell capacity; the reported radius is the smallest achievable toroidal distance at which the empirical cumulative field first reaches 0.8. The particle positions drawn here are schematic and are not measured data. (b) The event order, read directly from the kinetics module rather than from any description of it.

operator it errs by -18.74% in the static condition and -19.29% in the mobile one. Coarse-graining the lattice away is not available at this scale [Hughes, 1995, Spitzer, 1976].

Second, transport is capacity-limited. A move is refused when the destination cell is full, so the one-step transport factor is a function of the occupancy, which is itself the quantity being propagated. The equation for the mean density therefore does not close; we return to this in Section 3 and again in Section 8, because it is the principal thing this paper does *not* resolve [Liggett, 1985].

Third, the source has a history. When the source is mobile, every particle alive at a given instant was born somewhere along one shared trajectory. Their displacements are not independent draws around a fixed point: they share a common, autocorrelated origin process. Ignoring that shared history is the single largest error one can make in this system, as Section 6 shows quantitatively.

Because of these three features, the extent is a measured quantity in a model, and the methodological question becomes sharp: can it be *predicted*? Extents of this kind are ordinarily reported after the data exist, which makes it impossible to distinguish a model that predicted a number from one that accommodated it. The design below was built to remove that ambiguity, at the cost of committing publicly to two numbers and a tolerance before running anything.

2 Model and observable

The medium is a periodic square lattice of side $L = 36$ with a capacity of 16 particles per cell. A single source cell injects particles of the reported species; particles hop to nearest neighbours with probability $p_{\text{hop}} = 0.102633$ per step, subject to capacity, and die with probability 0.004 per step. Two conditions are compared: *static*, in which the source is immobilised, and *mobile*, in which it performs its own walk. Nothing else differs between them.

The event order within one step matters and was read from the kinetics module itself, not from a description of it: transport of each species in turn, then reaction, then decay, then feed and outflow

(Figure 1b). One consequence is used later: because reaction runs after transport, a particle born in step t takes no step in step t . Every one of the 141009 birth records examined across 40 arms occurred at the source cell exactly, with maximum coordinate offsets of 0 and 0; the point-source idealisation is not an assumption here but a checked property of the implementation.

Each run of the model is called an *arm* and is the unit of analysis throughout. An arm runs for 11000 steps, of which the first 2000 are discarded as burn-in, and the field is sampled every 50 steps thereafter, giving 180 frames per arm. The within-arm summary is the *median* of r_{80} over those frames; that rule was fixed by an earlier protocol and is inherited here unchanged, which turns out to matter a great deal (Section 5). Across arms we take the arithmetic mean. Frames, steps, particles and birth events are never treated as independent replicates for any confirmatory statement.

The burn-in is adequate for the mass but only marginally so for the shape. The mass relaxes with an e-folding time of 249.5 steps, so the burn-in is 8.01604 e-foldings and leaves a residual mass deficit of 0.000330124. The slowest shape mode on this torus has a time constant of 657.73 steps, so the same burn-in is only 3.04076 e-foldings of it and leaves a residual shape deficit of 0.0477986. That finite-time effect is not assumed away: it is carried inside the construction, which starts from an empty lattice and burns in for exactly the protocol duration.

3 Construction of the operator, and the freeze

3.1 The one-step conditional operator

Conditional on the state S_t , the expected occupancy vector obeys

$$\mathbb{E}[n(t+1) | S_t] = T(S_t)n(t) + b(S_t), \quad (1)$$

and this form is exact for this engine, not an approximation to it. Transport contributes a T that is a function of the occupancy, because the number of movers accepted out of a cell is the minimum of the number offered and the free capacity at the destination; reaction contributes a b that is likewise state-dependent, being the minimum of the available substrate and the free capacity at the source cell; and decay contributes an exactly linear factor $(1 - 0.004)$ with no state dependence at all. In the surviving record, births equalled that minimum in 0.866663 of steps, and the source cell had zero free capacity in 0.00104094 of them, measured over 57 arms.

Equation (1) does not survive taking expectations. Since T and n are both functions of the same occupancy, $\mathbb{E}[T(S)n] \neq \mathbb{E}[T(S)]\mathbb{E}[n]$, and the marginal density equation therefore does not close [van Kampen, 2007, Gardiner, 2009]. What *is* available is that the unblocked limit is a genuine closed linear operator, and that the departure from it is bounded: over a mean lifetime a particle is offered 25.5557 hops and expects 0.00908288 refusals, so 0.990917 of particles are never refused at all. Removing that fraction of moves, and concentrating every removal adversarially on outward moves, changes the reported radius by 0.01777%. The stationary profile therefore closes approximately, with a certified bound; the marginal density does not close at all. We record both, and we do not present the first as though it were the second.

FULL_ONE_STEP_CONDITIONAL_OPERATOR	CONDITIONAL_EXACT
STATIONARY_PROFILE_CLOSURE	APPROXIMATE_WITH_CERTIFIED_BOUNDS
MARGINAL_DENSITY_CLOSURE	NOT_CLOSED

Table 1: The three pre-declared endpoints. The predictions and the margin were fixed before any of these arms existed. *Strict pass* means the whole confidence interval, not only the point estimate, lies inside the margin.

Endpoint	Predicted	Observed	Deviation (%)	95 % interval (%)	Strict pass
Static absolute	6.0076	5.9991	−0.14	[−1.05, +0.77]	yes
Mobile absolute	8.0574	8.0771	0.24	[−1.18, +1.67]	yes
Mobile / static	1.3412	1.3464	0.39	[−1.30, +2.07]	yes

3.2 Terms measured, and terms dropped

Four candidate corrections were measured before the confirmation arms were designed, and three of them were dropped on the evidence rather than on judgement. The finite torus contributes nothing to the reported radius, because that radius is quantised to achievable lattice distances whose spacing near these values exceeds the correction. The finite horizon contributes 8.01604 e-foldings of mass relaxation and is carried inside the construction rather than corrected for. The intra-step order contributes 0.2008 %. Capacity rejection contributes 0.01777 %. Two terms were retained because they are large: the shared source trajectory, and the endogeneity of the birth flux. The birth flux is over-dispersed relative to Poisson, with a variance-to-mean ratio of 1.285, and the stationary population it implies, 120.845, agrees with the observed 120.569 to a ratio of 1.00229. The age distribution of the standing population is the geometric one [Feller, 1968] to within a ratio of 0.994456 across 6141 particles.

3.3 The equivalence margin, and what it is not

The tolerance was derived before any confirmation arm existed, from named terms, and was not chosen to accommodate anything. The model-error terms combine in quadrature to 0.6352 %; two sampling standard errors at 14 arms per condition amount to 2.219 %; the declared margin is 2.9 %. The point that matters is the referent [Wellek, 2010]: the margin is applied to the distance between the observation and the construction’s *point prediction*, not to the distance from zero. A margin built to contain the historical numbers instead would have had to be about twice as wide.

3.4 Freeze and inertness

Before any confirmation arm ran, the two absolute predictions and their ratio were written into a protocol file and the analysis modules were hashed; the frozen predictions carry the status `PRE_RUN` and the methods hash `FROZEN`. The apparatus that records the runs was checked separately: over 1500 steps an instrumented engine and a plain one reach a bit-identical state, hash `25b51d2a28943cc6497441fd81d8001be208094f909d1978ae325c903bbdaa86`, so `INSTRUMENTATION_ALTERS_THE_LAW = FALSE`. This is one paired run at one seed; it falsifies interference, it does not prove inertness for every trajectory.

4 Prospective confirmation

28 arms were then run, 14 per condition, on seeds drawn from a register that excludes every seed appearing anywhere in the historical record. All 28 were analysable: 0 extinctions, 0 technically invalid runs, and no protocol violation. Each arm recorded 44000 hop-ledger rows, 44000 source-substep rows and 11000 birth-substep rows.

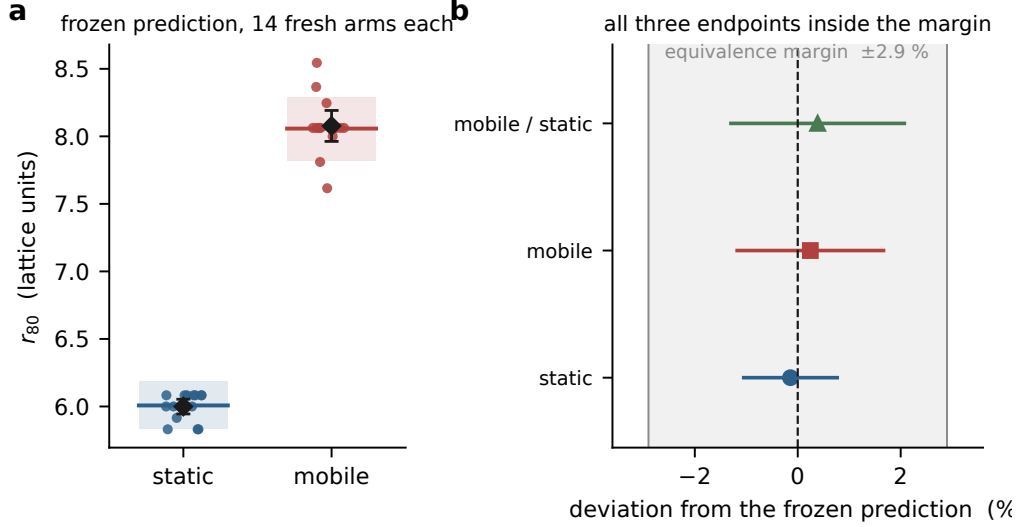


Figure 2: Prospective confirmation on 28 fresh arms. (a) The frozen prediction (line) with its equivalence margin (band) against the 14 arms of each condition (points) and their mean with a 95% interval (diamond). (b) The three pre-declared endpoints; every interval lies wholly inside the margin, and unity is excluded for the ratio.

Table 1 and Figure 2 give the result. The static prediction was met to -0.14% and the mobile one to $+0.24\%$, against a margin of 2.9% ; in both cases the entire confidence interval lies inside the margin, which is a stricter criterion than the one that was declared. The ratio of the two conditions was predicted at 1.3412 and observed at 1.3464 , a deviation of $+0.39\%$, and unity lies outside its interval: source mobility increases the extent, by a margin the construction anticipated.

What this would have looked like had it failed is worth stating, because it is what makes the agreement informative. Either absolute endpoint outside $\pm 2.9\%$, or a ratio interval containing unity, or a single extinction or technically invalid arm, would have been visible in exactly this table. The predictions were on the record beforehand; the arms were run once; the seeds were not replaced.

Two further observations bound what the agreement means. Capacity blocking averaged 0.0004 across the 28 arms, with a maximum of 0.000597 ; the full-capacity label in this work refers to the source term, not to a saturated lattice, and the regime tested here is not capacity-limited. And the qualification is bounded: it holds at one parameter point, on one lattice, for two mobility settings. The functional dependence on hop rate is not measured, only the contrast between the two settings that were frozen. The adjudicated disposition of the programme is `FULL_CAPACITY_SOURCE_RESPONSE_OPERATOR_QUALIFIED`.

5 The residual was in the summary rule, not in the field

The construction just tested was not built in a vacuum. An earlier record, made under the same frozen rules, had shown the reported radius falling *below* the ideal operator's value: -1.833% in the static condition and -5.172% in the mobile one, the latter over 116 arms at $z = -16.57$. A deficit that size, that consistent, and that clearly larger in the mobile condition reads like physics. It is not.

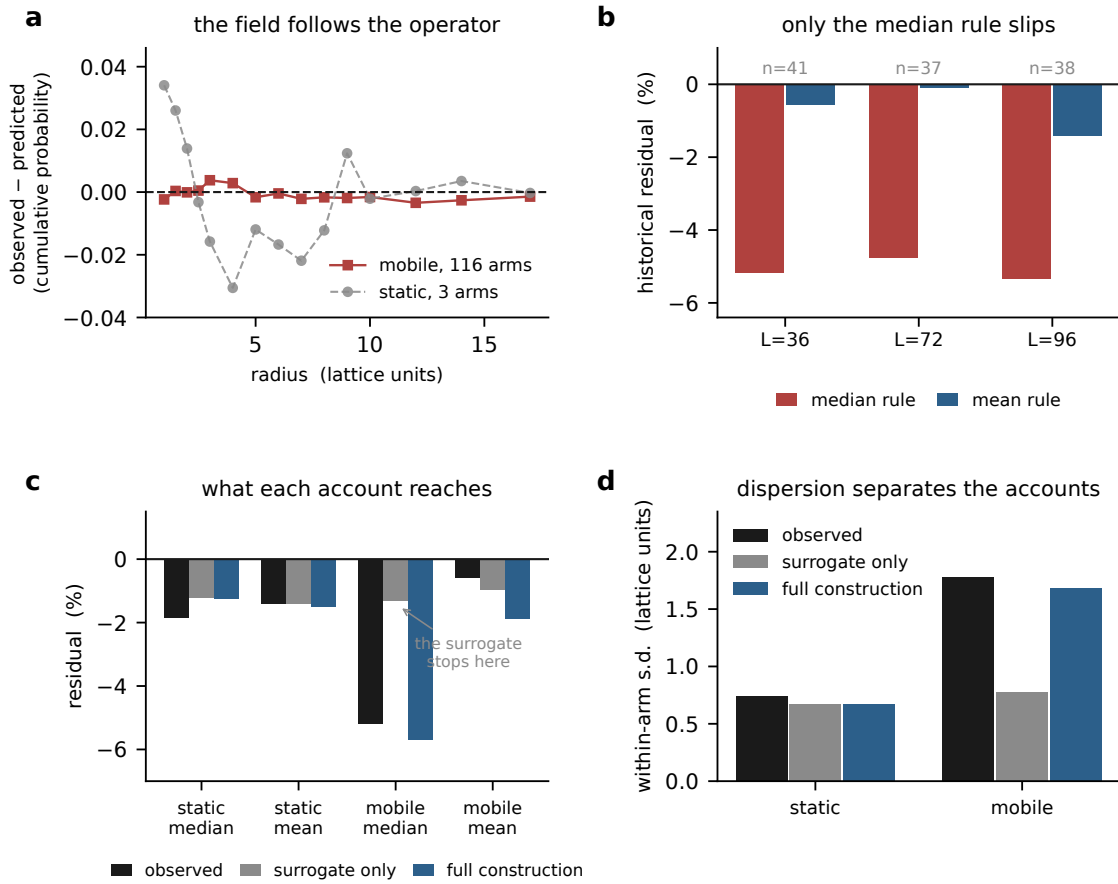


Figure 3: The summary rule, not the field. (a) Observed minus predicted cumulative radial profile at 15 radii. Over 116 historical mobile arms the largest standardised deviation is 0.64 and the largest difference 0.0038. The static curve rests on 3 arms and is shown for completeness only; see the caveat in the text. (b) The historical residual at three lattice sizes under the frozen median rule and under the mean rule, on the same frames. (c) Four summary cells under three accounts. (d) Within-arm dispersion under the same three accounts.

5.1 The field agrees at every radius

The first thing to check is whether the shortfall is in the field or in how the field is summarised. Over 116 historical mobile arms the observed cumulative radial profile follows the operator at every one of 15 radii: the largest standardised deviation is 0.64 and the largest difference in cumulative probability is 0.0038 (Figure 3a). The cloud is where the operator says it is. Only the summarised quantile is not.

The same comparison in the static condition is *not* informative and we do not use it. It rests on 3 arms. Its largest standardised deviation is 8.904, but that occurs at a radius where the absolute difference, 0.03407, is no larger than at radii whose standardised deviation is below one — a standard error estimated from three arms can collapse by chance, and here it did. The flag recorded in the source programme is scoped `MOBILE_ONLY` accordingly, and we report the static cell as not evaluated rather than quietly averaging it in.

5.2 A surrogate with no dynamics reproduces part of the deficit

Take the operator’s own stationary law, draw particles from it independently at the observed per-frame counts, and apply the identical summary rule. There is no lattice, no transport, no capacity, no source: only the estimator. That surrogate is already biased downwards, by -1.206% in the static condition and -1.307% in the mobile one, over 42 surrogate arms of 180 frames each. The mechanism is elementary and was flagged in the estimator’s own definition: r_{80} is a finite-sample quantile of a first crossing, which is a minimum over distance, and minima of noisy crossings are biased downwards [Serfling, 1980, Redner, 2001].

That accounts for most of the static deficit — -1.206% against -1.833% observed — and we say so no more strongly than that.

5.3 It does not account for the mobile deficit

In the mobile condition the surrogate reaches only -1.307% against -5.172% observed (Figure 3c). Roughly three quarters of the mobile deficit is not an estimator artefact in this sense, and something in the dynamics has to supply it. What supplies it is the source’s own wandering, quantified in Section 6: adding the shared trajectory moves the construction from -0.6563% to -4.424% , and adding the empirical birth flux takes it to -5.695% , against -5.172% observed. The construction slightly overshoots; its own Monte-Carlo standard error at that point is 0.6024% .

5.4 Dispersion orders the two accounts

Dispersion separates the accounts more cleanly than the central value does. The observed within-arm standard deviation in the mobile condition is 1.780; the surrogate produces 0.774, less than half of it; the full construction produces 1.681 (Figure 3d). On the fresh arms the same comparison was a pre-declared secondary control and was met: predicted 1.681 against 1.645 observed in the mobile condition, and 0.6667 against 0.6985 in the static. The mobile per-frame series is also right-skewed, at 1.072 against 0.313 in the static condition, which is what puts a median below its mean.

5.5 Changing the summary rule removes most of the discrepancy

On identical frames, the historical mobile deficit is -5.172% under the frozen median rule and -0.5752% under the mean rule (Figure 3b). The choice of summary rule, not the field, carries the bulk of it. The mean rule was also a pre-declared control on the fresh arms and was met: -1.24%

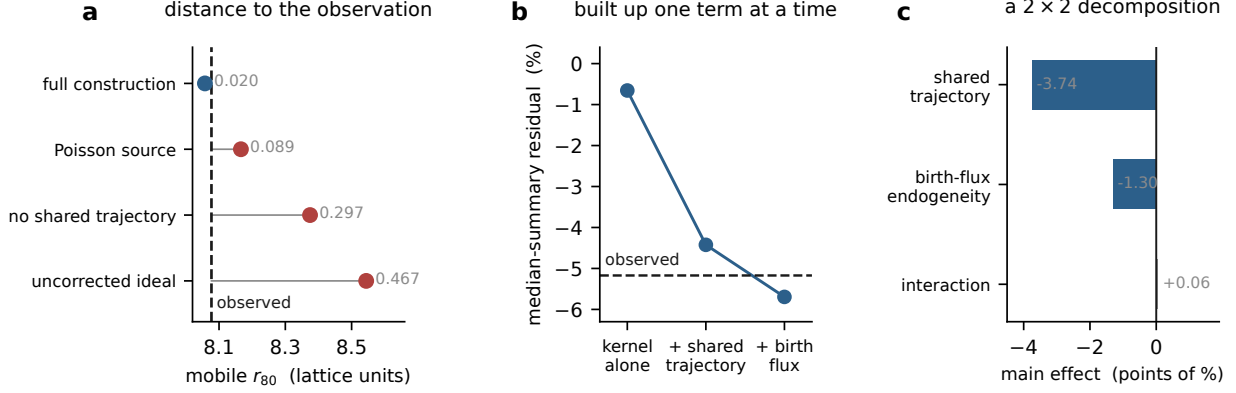


Figure 4: What the agreement depends on. (a) Distance between the observed mobile radius and four constructions. (b) The same constructions built up in physical order. (c) The two main effects and their interaction in a two-by-two decomposition.

observed against -1.485% predicted in the static condition, and -1.648% against -1.866% in the mobile one.

Two qualifications belong here rather than in the discussion. The mean rule is not unbiased — under the surrogate it is in fact slightly *worse* than the median in the static condition, -1.412% against -1.206% — it is merely less biased where it matters. And the construction’s largest single disagreement with the historical record is here: it predicts -1.866% for the mobile mean against -0.5752% observed, an overshoot of about a percentage point and a quarter. That cell is the weakest agreement in this paper and we do not present it as though it were among the strongest.

6 What the agreement depends on

An agreement is only informative if one can say what it would have taken to break it. We therefore removed each retained ingredient in turn and measured how far the prediction moved from the observed mobile radius, 8.0771. The full construction sits 0.0197 lattice units away. Replacing the endogenous birth flux by a Poisson source of the same mean moves it to 0.0889, about four and a half times further. Removing the shared source trajectory moves it to 0.2975, about fifteen times further. The uncorrected ideal value, 8.5440, is furthest of all (Figure 4a).

Built up in physical order rather than torn down (Figure 4b), the discrete kernel with a Poisson source gives -0.6563% ; adding the shared trajectory gains -3.767 percentage points; adding the empirical birth flux gains a further -1.271 . In a two-by-two design over the same two factors the main effects are -3.74 and -1.299 percentage points and the interaction is $+0.05543$, which is small enough that the two ingredients can be read additively here [Box et al., 2005, Cox, 1958]. We note the standing caveat on such decompositions: the two main effects and the interaction sum to the total by construction, so their sum is not independent evidence for either.

The reason the shared trajectory matters so much is visible in the dispersion. Removing it collapses the within-arm standard deviation from 1.681 to 0.7663 — that is, onto the static width — and the median penalty collapses with it. The mobile deficit is not a different physical regime; it is the same estimator acting on a wider per-frame distribution, and the distribution is wider because every particle alive at a given moment was born somewhere along one wandering path.

Two ingredients that might have been expected to matter do not. Capacity is one: blocking is 0.0004 on average and its adversarial effect on the reported radius is 0.01777%, two orders of

magnitude below the residual under study. The finite torus is the other: it leaves the reported radius unchanged at every size tested, because that radius is quantised more coarsely than the correction.

7 Project history, and what did not survive

This paper is one result of a longer programme, and honesty about the rest of that programme is part of reporting this one.

A line of work addressed to a different question — whether a region of the parameter space of this architecture has certain properties — was carried out in earlier sessions. It included a prospective calibration exercise and a subsequent arbitration over which route to take. The raw archives of that work did not survive the container resets during which this paper was written. Their narrative records exist outside this repository; their bytes do not. No quantity, figure, table or abstract sentence in this paper draws on any of it, and none of it is reconstructed from memory here.

What follows from that, stated plainly: whether the architecture admits a parameter region with the properties that line of work was asking about is **not tested** by this paper. The status is `NOT_TESTED`, which is a statement about the absence of a test and not about the answer. A reader will find no sentence anywhere in this paper asserting a negative answer to that question either; nothing here bears on it in either direction, and the evidence that might have borne on it is no longer available to be examined.

The historical arms used in Section 5 come from programmes whose raw archives are likewise not all present in this session. They are used descriptively — to characterise a record that motivated the analysis — and never as confirmatory replicates. Every confirmatory statement in this paper rests on the 28 fresh arms of Section 4, whose seeds, final state hashes and ledger sizes are listed in the supplement.

8 Discussion

Three things are established here, at one parameter point.

An operator built from the microscopic rules, with two named corrections and a margin derived from named error terms, predicted the stationary extent of a source-maintained field before the confirming data existed, and the data met it. The agreement is not confined to the summary statistic; the full radial profile follows the operator over 116 arms. Mobility increases the extent, by an amount the construction anticipated to within $+0.39\%$.

A residual that had every appearance of physics was largely an artefact of a summary rule. That diagnosis is worth more than the specific number attached to it, because the ingredients are generic: a first-crossing quantile is a minimum, minima of noisy crossings are biased downwards, right-skewed per-frame distributions put medians below means, and a wandering source widens the per-frame distribution without moving the underlying law. Any measurement that summarises a first-crossing length within a replicate and then averages across replicates is exposed to the same effect.

The agreement depends on ingredients that can be named and removed, and removing them degrades it by factors of four and fifteen. That is what distinguishes this from a fit.

What is not established is at least as important. The marginal density equation does not close, and nothing here closes it; the stationary profile closes only approximately, with a certified bound, and the reader should not read the agreement in Section 4 as an agreement of a closed theory. The qualification holds at one lattice size, one source strength and two mobility settings; the dependence on hop rate is a contrast, not a curve. The construction’s account of the mobile *mean* summary is

its weakest, overshooting by about a percentage point and a quarter. And the static radial profile comparison, on three arms, is not informative in either direction.

Status lines, repeated. H3_STATUS = NOT_TESTED; REPRODUCTION_STATUS = NOT_TESTED; HEREDITY_STATUS = NOT_TESTED; AUTONOMOUS_COHESION_STATUS = NOT_ESTABLISHED; X_LAWSPEC_BASELINE = UNCHANGED; ARCHITECTURE_CHANGE_NECESSITY = NOT_ESTABLISHED. No claim of biological reproduction, heredity, living status or autonomous cohesion is made anywhere in this paper, and no experiment reported here bears on any of them.

Data, code and provenance

Every number in this paper is emitted as a macro by `code/emit_numbers.py` from `provenance/PAPER_NUMERICAL_RECONCILIATION.json`, which records for each value its estimand, its unit of analysis, the source file it was read from, that file's SHA-256, the JSON path within it, and the sections in which it may appear. No number in this paper was taken from prose. The four figures are regenerated by `code/make_figures.py` and each ships a source-data file. The evidence tier of every programme cited is recorded in `provenance/PAPER_SOURCE_BINDING.json`, and what is missing, together with why, is recorded in `provenance/PAPER_MISSING_EVIDENCE_MATRIX.md`.

References

- George E. P. Box, J. Stuart Hunter, and William G. Hunter. *Statistics for Experimenters: Design, Innovation, and Discovery*. Wiley, Hoboken, 2 edition, 2005.
- D. R. Cox. *Planning of Experiments*. Wiley, New York, 1958.
- William Feller. *An Introduction to Probability Theory and Its Applications, Volume I*. Wiley, New York, 3 edition, 1968.
- Crispin Gardiner. *Stochastic Methods: A Handbook for the Natural and Social Sciences*. Springer, Berlin, 4 edition, 2009.
- Barry D. Hughes. *Random Walks and Random Environments, Volume 1: Random Walks*. Oxford University Press, Oxford, 1995.
- Thomas M. Liggett. *Interacting Particle Systems*. Springer, New York, 1985.
- Sidney Redner. *A Guide to First-Passage Processes*. Cambridge University Press, Cambridge, 2001.
- Robert J. Serfling. *Approximation Theorems of Mathematical Statistics*. Wiley, New York, 1980.
- Frank Spitzer. *Principles of Random Walk*. Springer, New York, 2 edition, 1976.
- N. G. van Kampen. *Stochastic Processes in Physics and Chemistry*. North-Holland, Amsterdam, 3 edition, 2007.
- Stefan Wellek. *Testing Statistical Hypotheses of Equivalence and Noninferiority*. Chapman and Hall/CRC, Boca Raton, 2 edition, 2010.